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Antifouling properties of amphiphilic poly(3-hydroxyalkanoate): an environmentally-friendly coating

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ABSTRACT

The development of biofouling is a major problem for marine industries. The conception of anti-fouling and fouling release coatings, with controlled physical-chemical properties is a promising strategy. Among them, amphiphilic systems, such as those composed of a hydrophobic polydimethylsiloxane matrix and a hydrophilic polyethyleneglycol additive are the most efficient and up to date. Despite their effectiveness, these systems are questioned due to the petrochemical origin of PDMS. The aim of this project was to substitute the PDMS matrix with a biopolymer, poly(3-hydroxybutyrate-co-3-hydroxyvalerate) and to improve its anti-adhesion properties through the elaboration of an amphiphilic system, via the addition of PEG or PHBHHx-*b*-PEG copolymer. The results, including the physico-chemical properties of PHBHV based coatings and static adhesion tests on a marine bacterium, *Bacillus* 4J6 and a diatom, *Phaeodactylum tricornutum* are compared with those of PDMS and PEG-modified PDMS coatings. Real antiadhesion activity was obtained for the PHBHV/PHBHHx-*b*-PEG system for a promising eco-friendly strategy.

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Introduction

Biofilm is the best lifestyle for microorganisms. It is defined as a well-organized structure of microorganisms living within an extracellular polymer matrix (Costerton et al. 1999; Flemming et al. 2016). The development of this biofilm is favored, on any biotic or abiotic surface, in contact with an aqueous medium. In the marine environment, the accumulation of living organisms (bacteria, diatoms, algae, invertebrates) is defined as biofouling. This phenomenon is causing serious problems for the use and durability of submerged materials. The massive colonization of ships' hulls induces an important loss of speed and long periods of fairing. This directly results in increased greenhouse gas emissions and economic cost (Piola et al. 2009; Nurioglu et al. 2015; Farkas et al. 2020). In aquaculture activities, the colonization of cages or nets modifies the waterline and clogs the meshes (Braithwaite and McEvoy 2005). Their regular renewal and cleaning procedures lead to

an additional cost for the professionals in the field (Silva et al. 2019).

In recent years, antifouling coatings (AF) have been developed to overcome these problems, particularly through the use of biocidal molecules. Their modes of action consist in destroying, repelling or blocking the adhesion and/or growth of organisms by chemical or biological mechanisms (UE 528/2012). Their use is nowadays highly controlled and regulated due to the environmental impact they cause (Antizar-Ladislao 2008). In this context, research has rapidly evolved towards the development of new materials free of any biocidal substances. Among the most promising AF coatings, hydrophobic coatings based on fluorocarbon (PTFE) and silicone (PDMS) have shown the best resistance against adhesion (Nendza 2007; Gabriel et al. 2016). One of the explanations could be related to their low surface energy as well as their low glass transition temperature (*T_g*). However, these coatings tend to adsorb proteins and accumulate diatom slimes (Sundaram et al. 2011). One strategy to

improve the performance of these anti-adhesion systems is the addition of a hydrophilic molecule within the matrix to obtain an amphiphilic coating. Very often, it is polyethylene glycol (PEG) or PEG derivatives that are among the most popular hydrophilic polymer used for that purpose (Yang et al. 2014; Ciriminna et al. 2015; Faÿ et al. 2016). A significant reduction in protein adsorption and diatom proliferation on PEG-silicone coatings has been observed (Wu and Hjort 2009; Camos Noguer et al. 2017a). The formation of a strong hydration layer that creates a steric barrier to adhesion is a possible mechanism to explain the role of PEG in amphiphilic systems (Morra 2000; Ngo and Grunlan 2017). In addition to their anti-adhesive properties due to their hydration layer and steric exclude volume effects (Ngo and Grunlan 2017), silicone-based amphiphilic coatings are known to be efficient fouling-release coatings (FRC). FRC's have a common characteristic of a low surface free energy. As a result, the attached organisms can be readily removed by the water shear force coming from mechanical cleaning or the ship's navigation (Hu et al. 2020).

Despite their efficiency, the disadvantage of these amphiphilic silicone-based coatings is their petrochemical origin. With the current concern of ecological transition, it seems unavoidable that these conventional polymers should be substituted by any bio-sourced polymers. One of these, poly(3-hydroxyalkanoate) (PHA), seems to be an ideal candidate for the development of these new eco-friendly coatings. Produced and stored in intracellular granules by numerous microorganisms, PHAs constitute a carbon reserve under conditions of energy stress. Their properties are based on their monomeric composition and molecular weight. The best-known PHA copolymer is poly(3-hydroxybutyrate-co-3-hydroxyvalerate) (PHBHV) consisting of the 3HB (3-hydroxybutyrate) and 3HV (3-hydroxyvalerate). PHAs are nontoxic, biodegradable and biocompatible with many cell types (Bugnicourt et al. 2014). The hydrolysis products of PHB, hydroxybutyric acid, is a natural constituent of blood and is therefore not toxic to humans (Anjum et al. 2016). PHA has thus become one of the most promising biomaterials used for medical applications (Koller 2018; Zhang et al. 2018; Elmowafy et al. 2019). The antibacterial properties of PHBHV have been studied, for example by Manecka et al. (2014), who modified the PHA surface with covalent grafting of acrylate monomers and observed a reduction of 90% and up to 99% in the adhesion of *Staphylococcus aureus* and *Escherichia coli* on the films compared

with raw PHBHV films. The anti-adhesion performance of PHBHV was also optimized by Poupart et al. (2015) by photografting 2-(methacryloyloxy)-ethyl trimethylammonium chloride (META) and an ampicillin-derived monomer onto PHBHV coatings. Bacterial adhesion and biofilm formation were also studied on two PHAs_{mcl}, poly(3-hydroxyoctanoate) (PHO) and poly(3-hydroxyundecanoate) (PHUA) by Mauclair et al. (2010). They showed biofilm formation to be lower or similar to that observed on the polystyrene coating used as a control. The bacterial adhesion properties of such PHA materials were also tested with *Vibrio campbelli*, a pathogenic bacterium from aquaculture, and the results showed the antibacterial activity of the short-chain fatty acid, β -hydroxybutyrate (Defoirdt et al. 2007).

In the present work, the adhesion of microorganisms on PHBHV and optimized amphiphilic PHBHV coatings was investigated under static conditions and compared with PEG enriched PDMS coatings used as control surfaces. A new PEG additives copolymer, poly[(3-hydroxybutyrate)-co-(3-hydroxyhexanoate)]-*b*-PEG (PHBHHx-*b*-PEG) synthesized *via* 'click chemistry' using photoactivated thiol-ene addition was used in the formulation of the amphiphilic PHBHV based coating. These new environmentally friendly coatings were tested against the adhesion of two types of microorganisms, known to be some of the major AF specimens in the marine environment: a marine bacterium, *Bacillus* 4J6, and a benthic diatom, *Phaeodactylum tricornutum* (Lejars et al. 2012; Zea-Obando et al. 2018).

Material and methods

Materials

Poly[(3-hydroxybutyrate)-co-(3-hydroxyvalerate)] (PHBHV) containing 30 mol % of 3-hydroxyvalerate was purchased from Pacific Biotech (Thailand). Poly[(3-hydroxybutyrate)-co-(3-hydroxyhexanoate)] (PHBHHx) with 20 mol% of HHx was supplied by CHEN Georges, Tshingua University, China. Polydimethylsiloxane (PDMS) was obtained from Momentive Performance Materials (France) as the trade mark RTV-615. O-(2-Mercaptoethyl)-O'-methylpolyethylene glycol ($M_n = 2000 \text{ g mol}^{-1}$) named 'PEGSH', 2,2-dimethoxy-2-phenylacetophenone (DMPA), allyl alcohol, dibutyltin dilaurate (DBTL), dichloromethane, Guillard's F/2 Marine Enrichment Basal Salt Mixture, NaCl, tryptone, antibiotics, yeast extract were purchased from Sigma Aldrich, Shanghai, China. A commercial PDMS-PEG

copolymer, traded under the name CoatOsil 7602, was obtained from Momentive Performance Materials (France). The copolymer has a molar mass of about 3000 g mol^{-1} and presents a grafted structure, as revealed by carbon and proton NMR. The copolymer contains 20–50 EO units per PDMS chain (Gillet et al. 2018).

Chloroform (CHCl_3), n-pentane, dimethylformamide (DMF) and dimethyl sulfoxide (DMSO) were purchased from Carlo-Erba. Dialysis membranes (6–8 kDa) were procured from Spectrum Labs (San Francisco, CA, USA). Syto 9 green dye was obtained from Thermo Fisher Scientific, Waltham, MA, USA. Glass microscope slides ($18 \times 18 \times 1 \text{ mm}$) were obtained from VWR International (Radnor, PE, USA).

Synthesis of PHBHHx-*b*-PEG copolymer

Unsaturated PHBHHx oligomer was obtained according to the procedure described by Ramier et al. (2012). PHBHHx (1g) was solubilized in anhydrous chloroform (10 ml) and $9.3 \cdot 10^{-3}$ moles of allyl alcohol and DBTL with a molar ratio $n_{\text{OH}}/n_{\text{DBTL}} = 34$ were added. The reaction was carried out in an Anton-Paar monowave 300, MW reactor. The MW source was a magnetron with a 2.5 GHz frequency (850 W power generator). The sample was heated at 160°C for 15 min. The oligomers were finally precipitated in 100 ml of n-pentane and dried under vacuum at 40°C (Yield 91%). Unsaturated PHBHHx was solubilized in DMSO (60 g l^{-1}). 2.5 eq molar of PEGSH was added in presence of 2 molar equivalents of DMPA. The mixture was irradiated at 100% for 30 min under agitation with the Hamamatsu lightening cure LC8 (L8251), equipped with a Mercury-Xenon lamp (200 W) coupled with a flexible light guide. The process from the addition of PEGSH and DMPA was repeated twice. The copolymer was then placed in 50 ml of DMF and purified twice by dialysis over 15 days by changing the water every 2 h.

Copolymer characterization

^1H NMR spectra in CDCl_3 were recorded at room temperature using a Bruker 400 MHz NMR spectrometer (Billerica, MA, USA). Average molar masses were determined by SEC experiments using Spectra Physics P100 pump and two PL-gel columns A Wyatt Technology Optilab rEX interferometric refractometer was used as the detector. Chloroform was chosen as the eluent at a flow rate of 1 ml min^{-1} and

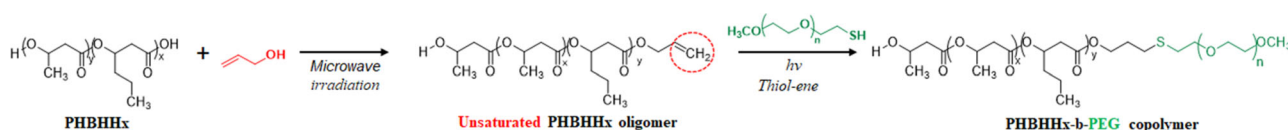
polystyrene standards were used for calibration. Thermal properties were determined using DSC-25 calorimeter from TA Instruments (New Castle, DE, USA) under nitrogen atmosphere. All the samples were analyzed between -70°C and 200°C with a heating rate of $10^\circ\text{C min}^{-1}$. After a rapid cooling from 200°C to -70°C (at $10^\circ \text{ min}^{-1}$) they were heated again to 200°C with a heating rate of $10^\circ\text{C min}^{-1}$. The melting temperature (T_m) and melting enthalpy (ΔH_m) were determined during the first heating run whereas the glass transition temperature (T_g) was determined at the second heating run. Structural characterization was obtained by X-Ray diffraction (XRD) using D8 Advance Bruker diffractometer (Billerica, MA, USA) with Cu $K\alpha$ radiation operated at 30 kV and 30 mA. Xray diffraction measurements were recorded over a 2θ range from 5 to 60° by a step 0.017° at an incident wavelength (λ) of $1.542 \text{ \AA}$.

Polymer surface preparation

PHBHV (10 wt.%) was dissolved for 2 h under reflux in dichloromethane. PEGSH or PHBHHx-*b*-PEG were added to the PHBHV solution, at a final concentration of 0.1 wt.% in solution (1% wt in dried coating) and were allowed to stir overnight. PDMS (50 wt.%) was dissolved for 2 h in dichloromethane. 0.1 wt.% of PEGSH or 0.01 wt.% of CoatOsil was added to the PDMS solution and allowed to stir overnight. A volume of $500 \mu\text{l}$ of each solution was then cast on a glass substratum (microscope slide). To avoid film shrinkage due to a rapid evaporation of the solvent, a polystyrene Petri dish cover was placed on top of each slide during the drying process that was set to 2 days for PHBHV coatings and 7 days for PDMS coatings at room temperature. The final coating thickness was estimated to be $\sim 100 \mu\text{m}$ as measured by an Elcometer 456 (UK).

Microorganism culture conditions

Bacillus (4J6) was isolated from the surface of glass covers immersed in Atlantic Ocean (Gulf of Morbihan, France, $47^\circ 34' 37'' \text{ N} - 2^\circ 44' 54'' \text{ W}$) for 6 h at 1 m depth. The marine bacterium was isolated on Zobell agar medium (4 g tryptone, 1 g yeast extract and 30 g sea salt) and then, incubated in Zobell medium for 24 h at 20°C with shaking. The cultures were allowed to produce a stock of several microtubes which was cryo-conserved in 20% glycerol at -80°C for the all experiments.



Scheme 1. Synthesis of PHBHHx-*b*-PEG copolymer.

The benthic diatom *Phaeodactylum tricornutum* (AC590) was obtained from the Algo-Bank (Biological Resource Center of the University of Caen Normandie, France). The diatom was grown in a sterile artificial seawater (SASW) culture medium with 2% Guillard's F/2 Marine Enrichment Basal Salt Mixture (stored at 4 °C before use), streptomycin (500 µg l⁻¹), penicillin G (1 g l⁻¹) and chloramphenicol (100 mg l⁻¹). Diatom suspensions were maintained under controlled illumination of 110 µmol photons m⁻² s⁻¹ white fluorescent lamps at 20 °C at 18 h/6h light/dark periods in a Hélios 600 phytotron (Cryotec, Saint-Gély-du-Fesc, France). After reaching the stationary phase (15 days), they were let in the dark for 40 h for synchronization.

Water contact angle

Water contact angles (θ s) were measured by a Digidrop (Digidrop, GBX, UK) at room temperature (23 °C, RH 50%). The volume of the water droplet tested was 3 µl. The analysis time was 3 s (above a phenomenon of evaporation is observed). Five different locations for each coated surface (18 × 18 × 1 mm) were chosen to obtain an average value.

Atomic force microscopy (AFM)

Images were recorded using the ScanAsyst® mode and the Peak Force Quantitative Nanomechanical Mode (PFQNM®) of a Multimode 8 AFM instrument equipped with a NanoScope V controller (Bruker, Santa Barbara, CA, USA). Topographical Images were recorded in the ScanAsyst® mode (Bruker, Santa Barbara, CA, USA) under ambient conditions (at 23 °C and RH of 50%), using a ScanAsyst-Air cantilever (resonance frequency of 70 KHz, spring constant of ~0.4 N m⁻¹ and tip radius of 10 nm) at a scan rate velocity below 2 µm s⁻¹. Topographic images were analyzed using the nanoscope Analysis software (V1.80). For each sample, a minimum of three different coated surfaces were prepared and observed and on each surface of a sample, a minimum a three different area were analyzed to ensure a good reproducibility in the sample preparation and surface topography.

PFQNM was used for determining the nanomechanical properties of the different samples. All measurements were performed under ambient conditions. In PFQNM, the piezo is vertically oscillating at a frequency of 2 KHz, with an amplitude of 150 nm. As the piezo scans the samples beneath the tip (scan velocity < 2 µm s⁻¹), a force-distance curves is recorded at each pixel, with a constant force that is maintained with the Peak Force feedback loop. Calibration of the two cantilevers' spring constant was obtained from the thermal tuning method for the ScanAsyst tip (spring constant of 0.34 N m⁻¹) or using the Sader method for the RTESP-525 tip (spring constant of 160 N m⁻¹). Their deflection sensitivities were determined from the linear part of an average of 10 force-distance curves obtained by ramping the tips onto a Sapphire surface. Their tip radius was calibrated using the relative method based on references samples (PFQNM SPM kit –12 M, Bruker), a PDMS standard of 2.5 MPa for the ScanAsyst tip and a Polystyrene film of 2.7 GPa for the RTESP-525 tip respectively. From these calibration standards, the peak force set-points were typically set to 300 pN for the PHA sample and to 20 pN for the PDMS surfaces.

The force curves from every interaction between the tip and the sample are analyzed in real time to provide nanomechanical map simultaneously with the topographical image. The indentation modulus is calculated according to the Derjaguin-Muller-Toporov (DMT) model (i.e. Hertz model including the effect of adhesion without any change in the contact area (Derjaguin et al. 1975). All AFM data analysis and image processing were made with NanoScope software version 1.8.

Adhesion test conditions and confocal laser scanning microscopy (CLSM)

Coated polymer films deposited on glass microscope slides (18 × 18 × 1 mm) were sterilized by UV radiation for 30 min, to ensure a non-contamination of the investigated coating before adhesion tests (Azemar et al. 2021). Contact angle measurements of the sterilized polymer films reveal that the wetting surface properties have not been modified after such UV treatment. For the bacterium, coated surfaces were incubated in 5 ml of a given suspension in solution of

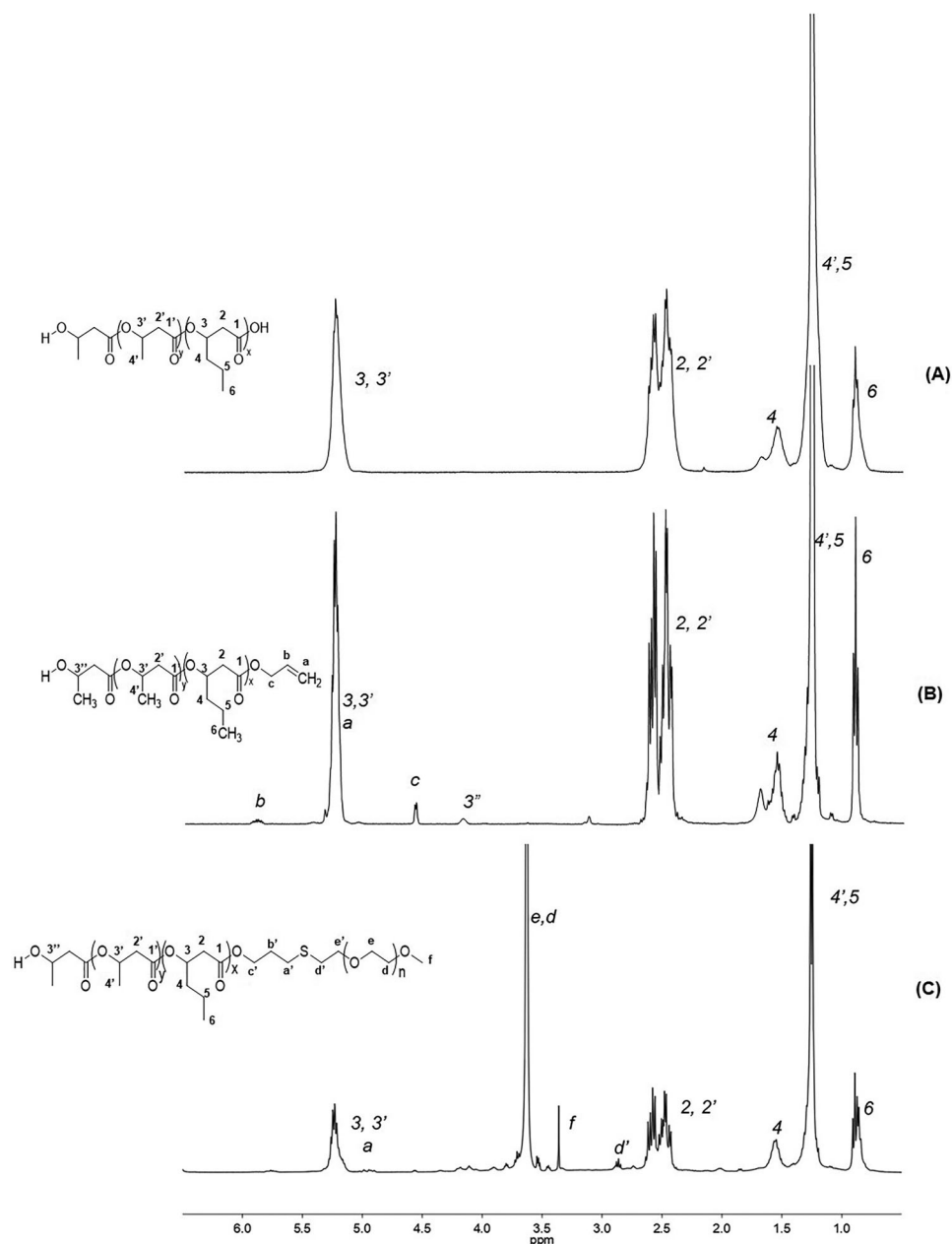


Figure 1. ^1H NMR of PHBHHx (A), PHBHHx oligomer (B), and PHBHHx-*b*-PEG (C).

SASW (10^8 cells ml^{-1}) for 2 h at room temperature under static conditions whereas for the diatom, coated surfaces were incubated in 5 ml of a given suspension in solution of SASW (10^5 cells ml^{-1}) for 72 h in a phytotron. Different adhesion times were tested in order to obtain a percentage of covering useful for image processing. Finally, an adhesion time of 2 h was chosen for the bacterium and 72 h for the diatom. After incubation, coatings were rinsed three times for 5 s in SASW solution to remove any non-absorbed microorganisms. Accumulated bacterial cells were stained with 5 μM Syto 9 green (485 nm excitation and 498–550 nm emission) for 20 min in the dark

before observation. The diatom cells were imaged *via* the autofluorescence of chlorophyll *a* (633 nm excitation and 650–700 nm emission). Images were captured with a confocal microscope LSM 710 from Zeiss (Zeiss, Germany). The percentage coverage was evaluated using Image J software (NIH, USA). The results reported are an average of a total of 30 independent measurements. Statistical analysis of the percent coverage data was performed with GraphPad Prism 8.2.1. *p*-values were calculated by t-test analysis ($p < 0.05$). Error bars represent the standard deviation (SD) between the fluorescence measurements of 40 randomly selected regions on five microscope slides

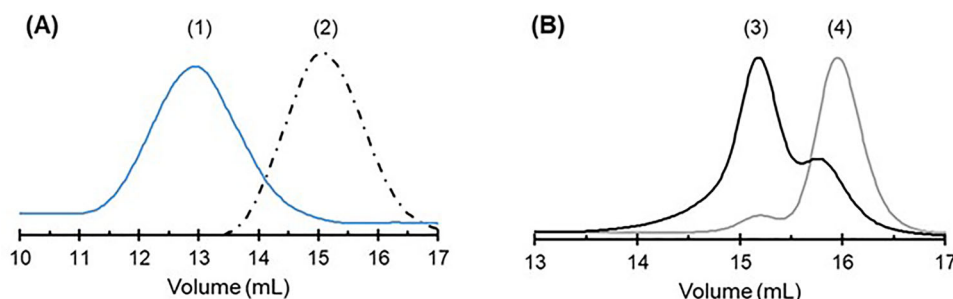


Figure 2. Chromatograms of PHBHHx polymer (A1), PHBHHx oligomer (A2) PHBHHx-b-PEG copolymer (B3), and PEGSH (B4).

(8 measurements per slide). The inhibition percentages were calculated for each coating according to the matrix reference surface.

Results

Synthesis and characterization of PHBHHx-b-PEG copolymer

The formation of PHBHHx oligomer having a double bond end-group occurred in one step by direct reaction in presence of allyl alcohol under microwave irradiation for 15 min (Scheme 1)

The presence of the unsaturated group on the PHBHHX oligomer was confirmed by ^1H NMR in CDCl_3 (Figure 1). The signals at 4.5 ppm and 5.8 ppm are characteristic of methylene (H_c) and methine (H_b) of the unsaturated terminal group. The molar mass of the oligomer determined by the ratio of the integrations of the signals at 2.5 ppm (2, 2') characteristic of CH_2 and at 5.8 ppm = CH is about $5\,600\text{ g mol}^{-1}$. Then the PHBHHx-b-PEG copolymer was synthesized by a thiol-ene reaction using thiolated PEG in the presence of DMPA as photoinitiator. The molar masses were also determined by SEC (Figure 2, Table 1). The molar mass of the oligomers determined by SEC is slightly higher than the molar mass determined by NMR that was previously observed in the case of polyhydroxyalkanoates because the molar mass is determined with PS standards. The molar mass of the copolymer PHBHHx-b-PEG is more important than the molar mass of the oligomer attesting of the formation of the copolymer. Nevertheless, the chromatograms showed the presence of free PEG even after purification by dialysis. The spectrum of PHBHHx-b-PEG revealed the presence of methylene and methyl groups of PEG at 3.6 ppm ($\text{H}_{e,d}$) and 3.4 ppm (H_f) respectively (Figure 1C). The disappearance of the unsaturated end group of the PHBHHx oligomers confirmed that there were no more PHBHHx free oligomers. The interaction between the free PEGSH and the copolymer seemed to be strong

enough to keep some PEGSH entrapped in the copolymer. Consequently, the dialysis process is not sufficient to remove all the free PEG, which will interact with the PEG block of the copolymer which has the particularity of being amphiphilic. By comparison with the integrations of the H_f peak at 3.4 ppm and 2, 2' at 2.5 ppm, the free PEG was about 30% w/w. The glass transition temperature of the copolymer was measured at -10.4°C (Table 1) and the crystalline structure was confirmed by X-Ray diffraction analysis (Figure 3). The PHBHHx oligomer is semi-crystalline with an orthorhombic unit in a helical form with space group $\text{P}2_12_12_1$ (Sato et al. 2004) with $a = 5.68\text{ \AA}$, $b = 13.03\text{ \AA}$ and $c = 5.90\text{ \AA}$ (Sato et al. 2004). As PEGSH is also a semi-crystalline polymer, the PHBHHx-b-PEG copolymer showed the crystalline structures of the two blocks.

Polymer surface preparation

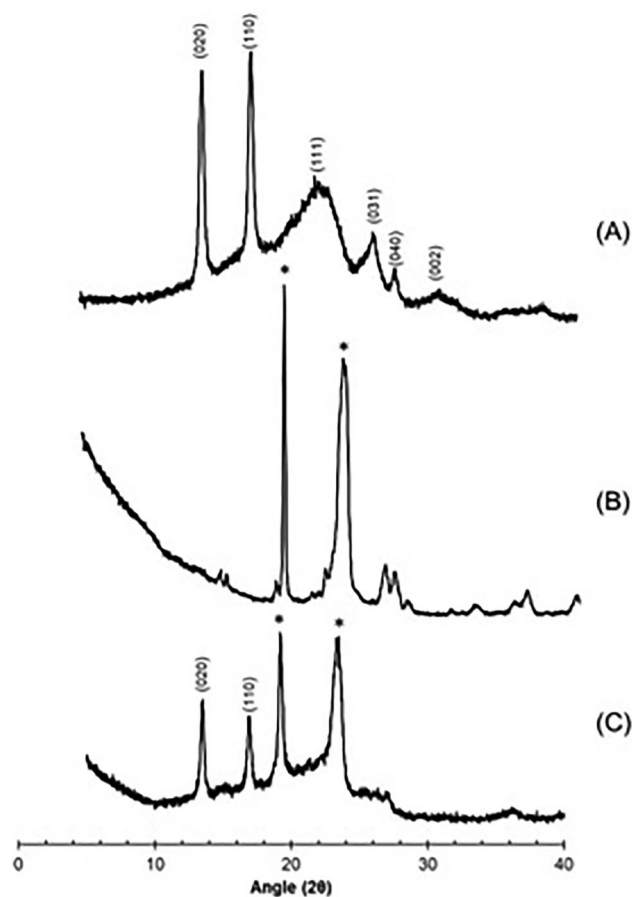
Six different coatings were studied in this work: three PDMS coatings (PDMS, PDMS/PEG and PDMS/PDMS-PEG) and three PHBHV coatings (PHBHV, PHBHV/PEG and PHBHV/PHBHHx-PEG). The same proportion, in weight, of PEG (free or in its copolymer form) was introduced into each modified coating, as shown in Table 2.

Water contact angle

At first, the hydrophobicity/hydrophilicity of the different coatings were investigated by means of water contact angle measurements. The results are shown in Figure 4. It was observed that the PHBHV coatings, with a contact angle of 87° , were slightly less hydrophobic than the PDMS coatings (contact angle of 95°). These measurements are in good agreement with the literature data for both polymers (Insomphun et al. 2017; Gillet et al. 2018). For the PHBHV coatings, addition of PEG, either in the free form or in its copolymer form, reduces the water contact angle of about 10° . No statistical difference

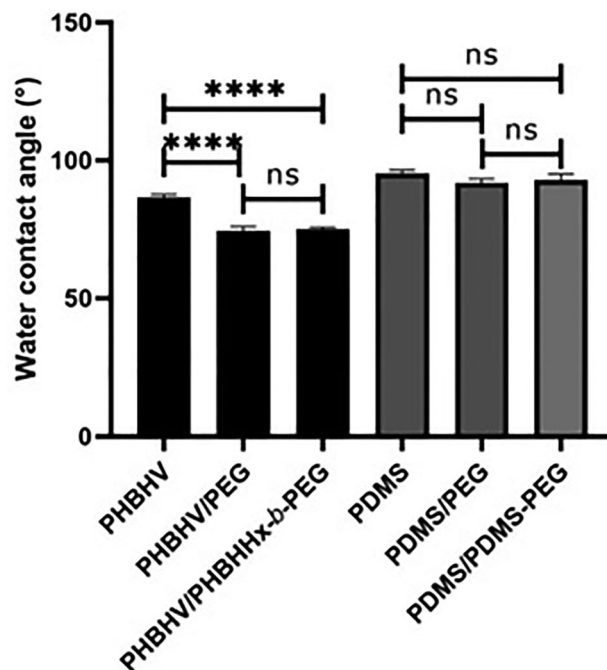
Table 1. Characteristics of PHBHHx alkene-terminated, PEGSH and PHBHHx-b-PEG copolymer determined by ^1H NMR^a, SEC^b and DSC.^c

	Mn (g mol ⁻¹) ^a	Mn (g mol ⁻¹) ^b	Ip ^b	Tg (°C) ^c	Tm (°C) ^c	ΔH (J g ⁻¹) ^c
PHBHHx	5600	7 200	1.7	-10.8	91.0	51.1
PEGSH	2130	3 750	1.2	Not detected	54.0	178.3
PHBHHx-b-PEG	–	9 230	1.6	-10.4	91.2	11.8
					52.1	84.0

**Figure 3.** DRX spectra of PHBHHx alkene-terminated (A), PEGSH (B), PHBHHx-b-PEG copolymer (C).**Table 2.** Blend codification and compositions.

Sample code	PHBHV (wt%)	PEGSH (wt%)	PHBHHx-b-PEG (wt%)
PHBHV	10	–	–
PHBHV/PEG	10	0.3	–
PHBHV/PHBHHx-PEG	10	–	0.1
Sample code	PDMS (wt%)	PEGSH (wt%)	CoatOsil (wt%)
PDMS	50	–	–
PDMS/PEG	50	0.1	–
PDMS/PDMS-PEG	50	–	0.01

between the water contact angle of the two PEGylated PHBHV coatings can be noted. In comparison with the PDMS coatings, the addition of free PEG or copolymer PDMS-PEG, did not statistically modify the water contact angle, compared with the native PDMS film. It should be remembered here that the

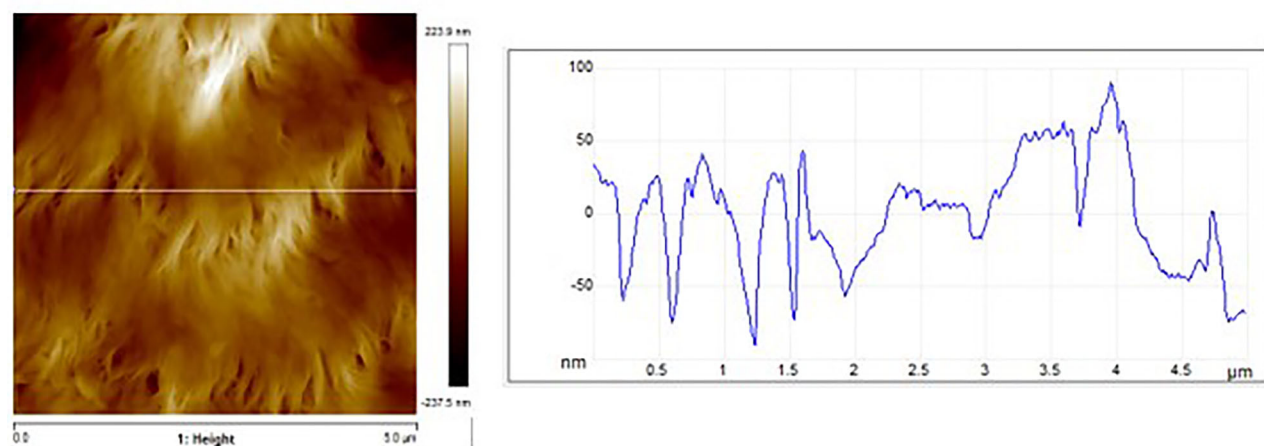
**Figure 4.** Water contact angle measurements of PHBHV and PDMS films measured by the sessile droplet method. $n = 20$. **** indicates significant differences ($p < 0.0001$) and ns indicates not significant differences between measurements.

water contact angles (WCA) were recorded within 3 s, to avoid any evaporation of the sessile drop.

Atomic force microscopy

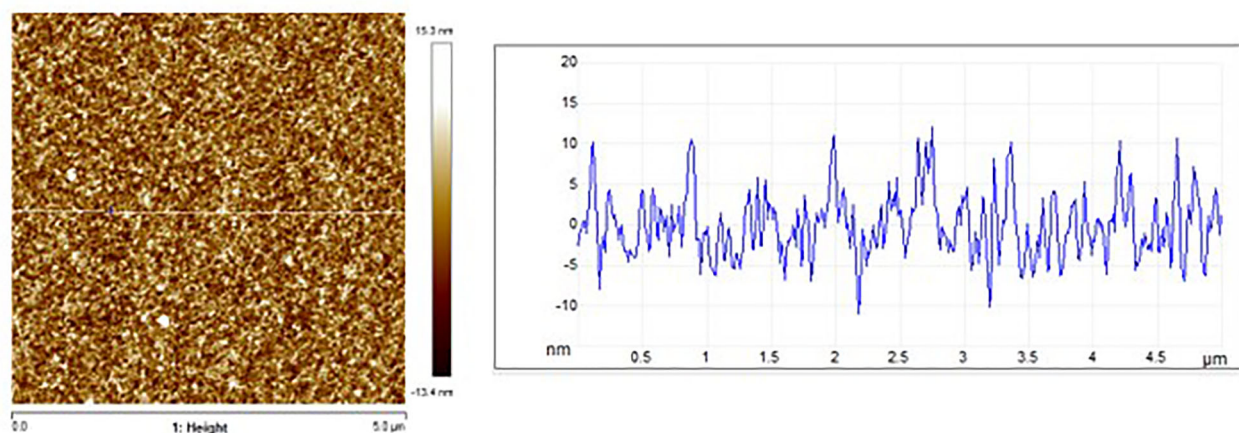
A section analysis of both polymer coating and the resulting roughness (RMS), maximum height (Z max) and projected surface area (Sa) data are listed in Figure 5. The PHBHV coatings present a roughness around 25 nm at $1\mu\text{m}^2$ and up to 100 nm at larger area of $100\mu\text{m}^2$ whereas PDMS surfaces show RMS around 4 to 5 nm, even at large scale of $100\mu\text{m}^2$. The results clearly show strong differences within the surface topographies of the two coatings. On the one hand, the PDMS coatings were smooth, showing a typical mesh-like network structure (Figure 6B) with pore sizes roughly estimated to be between 15 and 18 nm, in good agreement with the literature values (Drebezghova et al. 2020). On the other hand, the PHBHV coatings showed different features. The AFM images, at large scale (10 μm or more) reveal that the

A)



	RMS (nm)	<u>Zmax</u> (nm)	S _A (μm ²)
1μm ²	25.2 ± 5.6	171.2 ± 37.9	1.2 ± 0.1
25μm ²	71.4 ± 7.9	476.0 ± 95.6	28.1 ± 0.5
100μm ²	98.6 ± 19.1	934.2 ± 254.8	111 ± 1.5

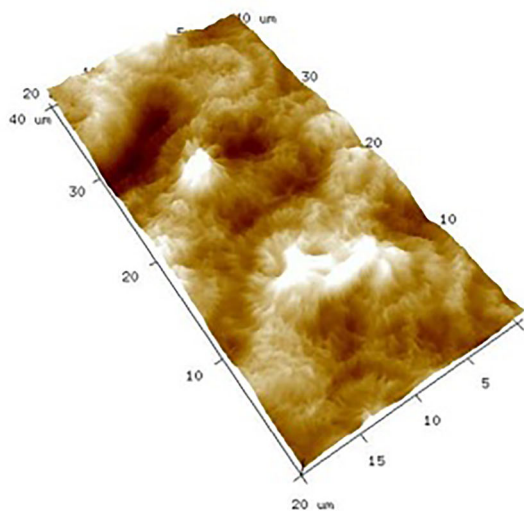
B)



	RMS (nm)	<u>Zmax</u> (nm)	S _A (μm ²)
1μm ²	4.0 ± 1.1	40.0 ± 2.0	1.3 ± 0.1
25μm ²	4.5 ± 0.2	40.0 ± 1.8	26.8 ± 0.2
100μm ²	4.2 ± 0.2	40.0 ± 0.3	102.7 ± 0.6

Figure 5. Representative AFM images (5 μm x 5 μm) and corresponding section analysis for: (A) PHBHV film and (B) PDMS film. The section analysis of both images shows the difference in height between the crystallized PHBHV film and the amorphous PDMS film. Topographical characteristics are given in the table below for three different scan areas in correlation with the size of the adsorbed organisms. RMS represents the root mean square roughness, Zmax is the maximum height on the image and S_A is the projected area of the surface.

A)



B)

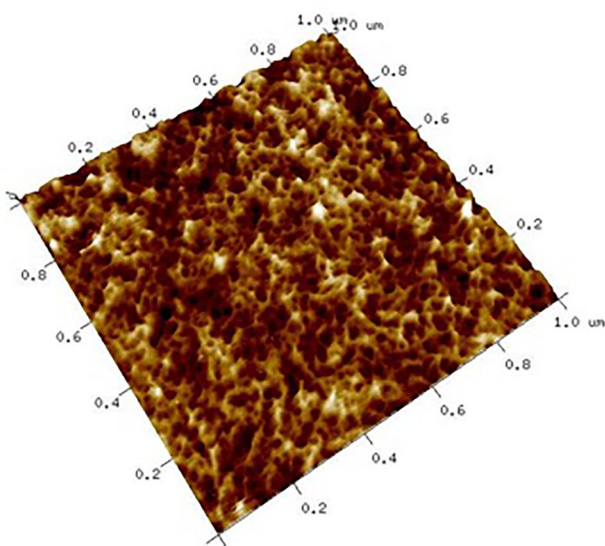


Figure 6. (A) Typical AFM height images of PHBHV films at larger scale (40 x 20 μm) showing spherulites and crystallite structures and (B) AFM height image of the PDMS film (1 μm x 1 μm) showing the mesh-like structure.

macromolecules were organized in large concentric spherulites (Figure 6A), as described by Nguyen Tri and Prud'homme (2018). The semi-crystalline state of the coating was also confirmed by DSC measurement showing a fusion peak at around 171 $^{\circ}\text{C}$. The enthalpy of fusion is about 100 J g^{-1} , indicating that the coating contained about 65% of crystallites (Barham et al.

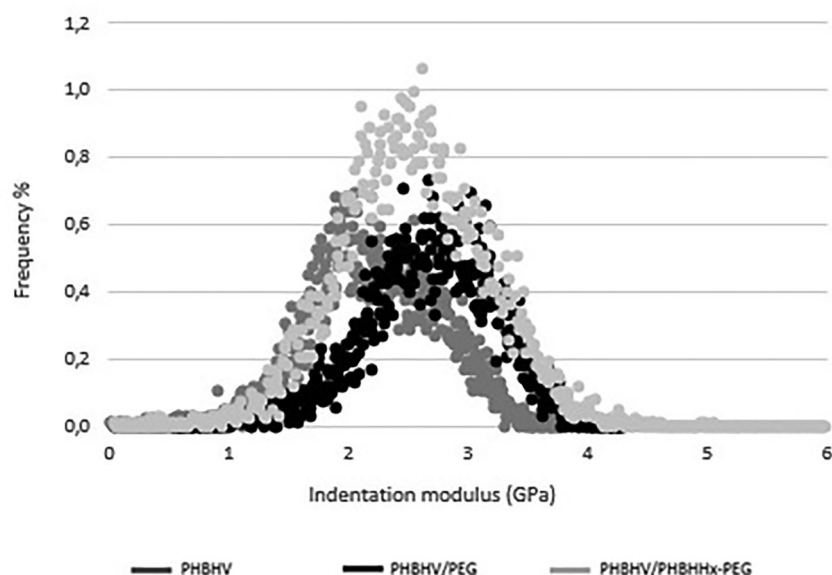
1984). From the AFM images, it can be deduced that the diameter of the spherulites varied between 15 and 20 μm , with interspacing from ridge to ridge of about 2 μm . Between these ridges, the PHBHV macromolecules were organized in long and large flat on lamellar crystallites, with branches growing in a radial direction (Figure 5). Using combined AFM-IR technique, Nguyen Tri and Prud'homme (2018) reported that PEG may form nanodomains of segregated particles preferentially located in the interfibrils of PHBHV stacks. The data for the coatings, containing free PEG or PEG in its copolymer form, are similar to those of the raw polymer coatings, indicating that the addition of PEG did not influence the surface topography of the coatings, at these scales and when observed in their dry state and under ambient conditions. The images of the different films, at a scale of 5 μm X 5 μm are available.

The nanomechanical properties of the different films were investigated using the PFQNM technique, in their dry state. The results, presented in Figure 7 show an important difference between the indentation modulus of PHBHV and PDMS coatings, with more than 1000 orders of magnitude. On the one hand, the PDMS coatings had an indentation modulus around 1 MPa, in good agreement with the literature data (Gökaltun et al. 2019). On the other hand, an indentation modulus of between 2 and 3 GPa was found for the PHBHV coatings. Addition of PEG, at this concentration, did not change the indentation modulus of the PHBHV analogs coatings. For the PDMS coatings, only the coating doped with the PEG copolymer presented a small shift toward a lower modulus, compared with the native PDMS coating.

Bacterial and diatom adhesion

At first, it was observed that native the PHBHV and the native PDMS coatings showed similar percentage coverage of the marine bacterium, at 0.6% (Figure 8). The addition of amphiphilic molecules, which was expected to improve the anti-adhesion properties of the coatings, can only be observed for the films doped with PHBHHx-PEG in the copolymer form. While no statistical difference in the percentage bacterial coverage between the native PHBHV and PHBHV/PEG coatings could be observed (0.7%), an important decrease of 50% in the amount of adhered bacterial cells is observed for the films containing the PHBHHx-PEG in the copolymer form. For the PDMS coatings, the trends observed are similar those of the PHBHV coatings, with a non-significant anti-

A)



B)

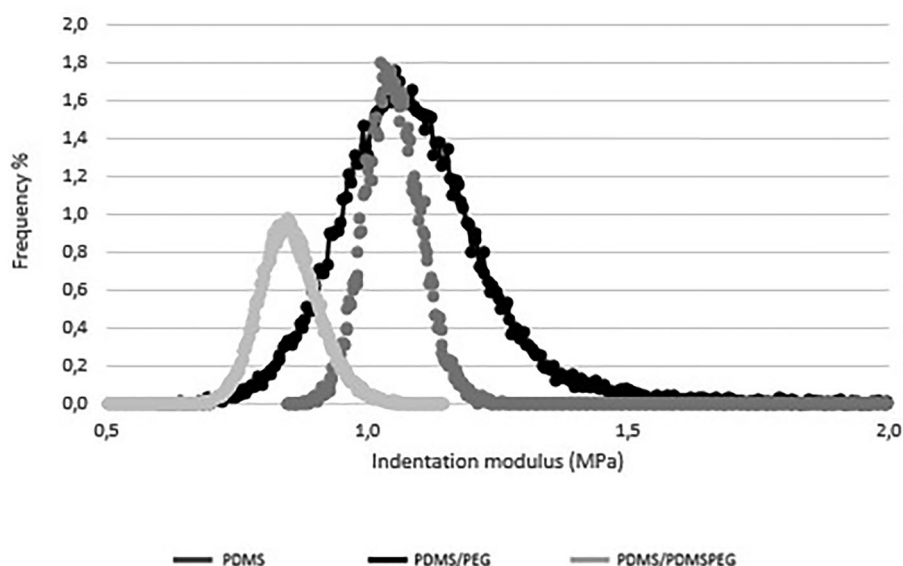


Figure 7. Distribution of the indentation modulus for (A) PHBHV coatings and (B) PDMS coatings.

adhesion effect for the PDMS/PEG coatings and an important decrease in the amount of bacterial cells adhered on the PDMS/PDMS-PEG coatings. For this latter coating, the adhesion was even reduced by 85% compared with the native PDMS coatings (Figure 8). The anti-adhesion properties of the authors' coatings was also tested with the diatom

(Figure 9). Native PHBHV and PDMS coatings show a similar efficiency (no statistical differences) against adhesion of the diatom cells. For the PHBHV coatings, the addition of PEG in free form or as a copolymer significantly reduced the adhesion of the diatom cells. Surprisingly, a similar behavior was not observed for the PDMS coatings since the addition

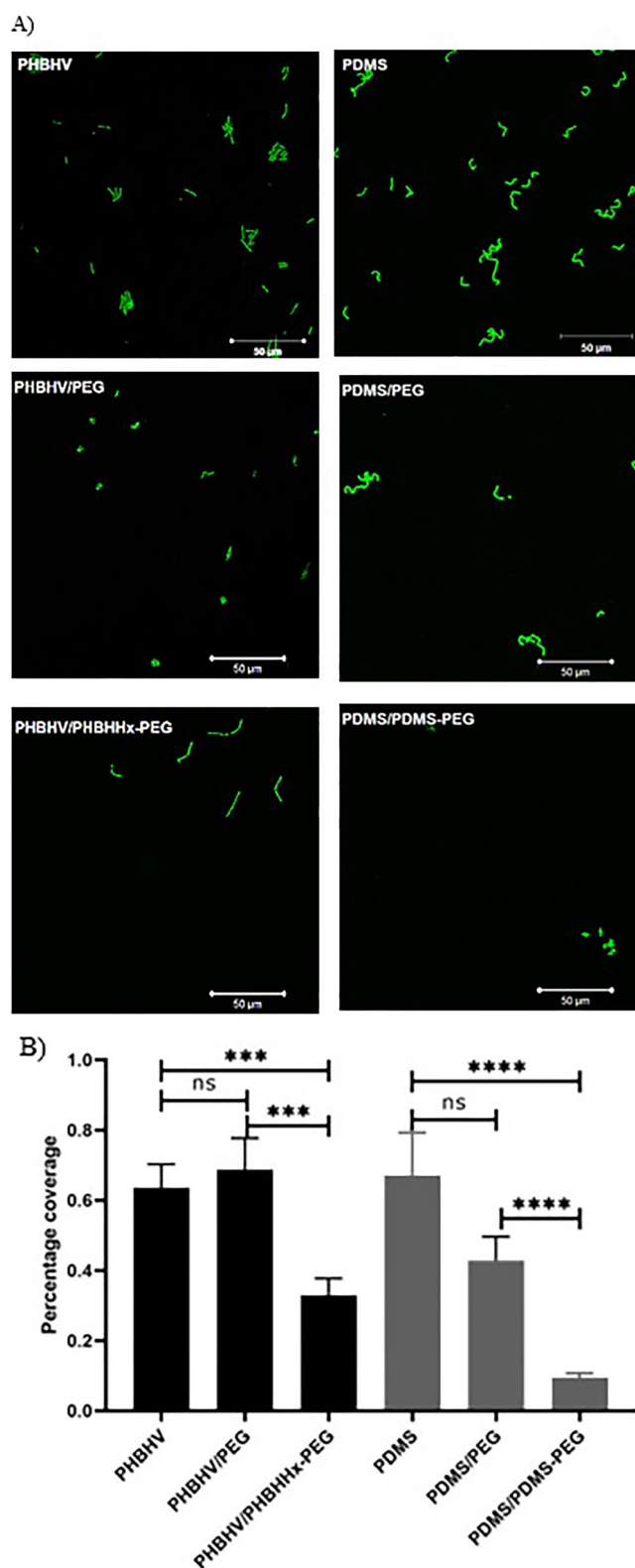


Figure 8. Settlement of the bacterium *Bacillus* (4J6) ($t = 2$ h; initial concentration = 10^8 cells ml^{-1}) on PHBHV and PDMS films. (A) CLSM images, (B) percentage coverage. $n = 30$. **** indicates significant differences at $p < 0.0001$ and *** $p < 0.001$ and ns indicates not significant differences.

of free PEGSH increased the adhesion. Finally, for the PDMS/PDMS-PEG coatings, the amount of diatom cells on the surface decreased by 96% compared

with the pure PDMS, thus reaching a similar low organism coverage as that obtained with the marine bacterium.

Discussion

The design of non-adhesive and fouling-release surfaces (FRC), requires control of the physico-chemical properties of the coatings (Lejars et al. 2012; Pistone et al. 2021). Wettability, surface topography and stiffness are identified parameters known to play a role in the anti-adhesion or in the release properties of the coating. For example, when considering hydrophobic coatings, their low surface energy eliminates the possibility of strong polar interactions with biomolecules or microorganisms. However, hydrophobic surfaces are not specifically considered as efficient AF surfaces. Furthermore, with low interactions on such hydrophobic surfaces, the elimination of fouling is rather easily enhanced under hydrodynamic conditions, thus suitable for fouling release applications. On the contrary, considering hydrophilic coatings, with high surface energies, their strong affinity for water molecules removes any thermodynamic advantage from the adsorption of biomolecules or microorganisms (Pistone et al. 2021). Current research builds on these concepts and deals with new ideas of surface composition that inhibits, at the same time, fouling and fouling release properties. Recent concepts focusing on surface composition include amphiphilic surfaces including PDMS-PEG systems (Camos Nogueir et al. 2017b).

The first parameter to be investigated is the surface chemistry of the coatings. With an angle of $\sim 90^\circ$, PHBHV coatings can be considered as hydrophobic coatings but remain less hydrophobic than PDMS coatings. Simply considering this difference in hydrophobicity, it is found that the adhesion of the bacterium and the diatom was more or less similar between the two coatings. Optimization of the anti-adhesive properties of PHBHV and PDMS was performed by adding PEG (in free form or copolymer). For the PHBHV coating, the addition of PEG to both forms allowed a decrease in the hydrophobic character on the surface of the coating. This result was not found for the PDMS coatings since the addition of PEG did not decrease the hydrophobic character of the surface, according to the experimental WCA measurements. A possible explanation for this result could come from the short analysis time. Contact angles with water were measured after 3 s, since above 10 s, evaporation of the drop, that can be observed, affected the measurement. The migration of PEG within the PDMS elastomer network is an important issue to ensure good stability and anti-adhesion properties of the coating over a sufficient period of time (Hawkins et al. 2014). It is, for example, proposed in the literature

that during solvent evaporation, a possible phase segregation of PEG can drive most of it to the polymer-air interface (Galli and Martinelli 2017). A longer time of immersion of such surfaces may induce polymer reorganization inside the coating, with consequences on both the chemical surface properties and the surface topography. This surface modification for an amphiphilic polyethylene oxide silane polymer in a silicone matrix has been observed by AFM *in situ* measurements (Hawkins et al. 2014). These authors reported that the surface topography of the amphiphilic silicone increased by a factor of 3 after immersion for 60 min in an aqueous medium as a result of polymer reorganization. Clearly, in the present experiments, such surface modification was not directly observed, either by WCA or by AFM, since the AFM studies were not conducted under liquid conditions. However, independently of this possible polymer reorganization, PEGylation of PDMS or PHBHV films, especially using the PEG copolymer form, provides a better anti-adhesion performance over a short period of immersion, as revealed by adhesion tests. These observations clearly highlight the important role of PEG in the hydrophobic coatings and the effectiveness of amphiphilic polymer surfaces in reducing the marine fouling process.

The second parameter that was analyzed was the surface topography. This parameter is relatively controversial in the literature. When considering AF surface coatings, it is for example admitted that a decrease in the surface roughness could act against bacterial adhesion, by reducing the anchoring area (Truong et al. 2010). Ludecke et al. (2016) did not consider the roughness (RMS) as a pertinent parameter, but they rather considered the peak density on their surface as playing a major role in the adhesion process. The presence of irregularities on the surface like peaks is said to favor the number of contact points between the microorganisms and the surface. Thus, if considering that the number of peaks decreases, adhesion can be consequently limited due to the decrease in the number of possible contact points. Of course, this statement is size dependent, for both the surface topography and the adhering organisms.

Opposite results have also been found by some authors (Singh et al. 2011; Liu et al. 2016). For example, Liu et al. (2016) observed that increased nanoscale roughness on PDMS coatings inhibited adhesion of Gram-positive and Gram-negative bacteria. They postulated that the increase in RMS and associated wetting energy contributed to enhancing

some protein adhesion responsible for decreasing bacterial attachment on such surfaces, acting like a passivation layer. Finally, when considering fouling release coatings, hydrodynamic forces and shear stress on the coating can be responsible for fouling release (Brady and Singer 2000). Under such conditions, the smoother the coating is, the most important can be the released microorganisms (Azemar et al. 2021).

In the present case, following the adhesion tests performed under static conditions, it is difficult to find a simple explanation for the role of the roughness of the coatings. In a first approximation, it is possible to think that the higher RMS of the PHBHV based films in contrast with the PDMS based coatings should favor the adhesion of microorganisms if simply considering the corresponding increase in the surface area. However, this may not be the unique explanation. The fact that adhesion tests with a bacterium and a diatom on the PEG copolymer enriched PHBHV come close to the performance of PEG doped PDMS coatings, despite a higher RMS, minimizes the role of this criterion to explain its good AF performance.

Surface stiffness is another factor that has been recently cited impacting microorganism adhesion but the exact link between the film's mechanical properties and its adhesion properties is still under debate. For example, Arias et al. (2021) found that the adhesion of *E. coli* (24 h under static conditions) and its biofilm formation are independent of the substratum stiffness of PDMS coatings in the range of 0.02 to 2 MPa. In another study, Kolewe et al. (2015) observed that fewer bacteria adhere to softer hydrogels (adhesion time of 2 h and 24 h under static conditions) and suggested that material stiffness could reduce the initial adhesion of bacteria. In the present study, the stiffness of the PHBHV coatings was much higher than that for the silicone coatings (Figure 6). However, understanding the role of the stiffness of the PHBHV coatings is complicated since in regards to the reference coating (PDMS/PDMS-PEG), all the physical parameters measured, including the indentation modulus, the topography and the wetting properties varied. Nonetheless, by simply comparing the three PHBHV coatings, with and without PEG, the adhesion of microorganisms was reduced when PEG was added, whereas the indentation modulus and the RMS were the same. It can be deduced from this observation that for PHBHV coatings in that range of modulus, the wettability parameter and hence its amphiphilic character is the important parameter that governs the AF properties. Similar to the behavior of

PHBHV coatings, the addition of PEG (in the copolymer form) in PDMS resulted in an important decrease in the surface coverage of the bacterium and the diatom. This result was not found for the PDMS/PEG system since on the contrary the addition of free PEGSH significantly increased adhesion of the diatom. One of the explanations could be that due to a probable leaching of PEGSH during the 72 h immersion, a modification of the surface topography occurred, thus favoring the adhesion of the diatom. It should be noted that the amount of diatom cells adhered on these systems was three times higher than the amount of adhered bacterial cells, which may be indicative of the complex nature of the forces involved in the adhesion mechanisms of microorganisms in the marine environment and their interaction with the contact surface. However, the difference in the adhesion time (2 h for the bacterium and 72 h for the diatom) can also influence the amount of adhered microorganisms. While addition of an amphiphilic polymer to the PDMS hydrophobic coating has demonstrated efficient AF properties towards a bacterium and a diatom, PEG modified PHBHV films also show promising anti-adhesion properties for some marine microorganisms. Despite a GPa range indentation modulus and a fairly high RMS due to its semi-crystalline structure, the association of an amphiphilic polymer within the hydrophobic PHBHV polymer matrix allows the design of interesting candidates for ecofriendly AF coatings.

Conclusions

This study explores the possibility of using the polymer poly(3-hydroxyalkanoate) as an alternative to the more conventional PDMS coatings and optimizing its AF properties *via* the design of amphiphilic coatings. The advantage of this substitution of the silicone hydrophobic matrix is to progressively replace a petrochemical polymer (PDMS), responsible for environmental pollution with a biodegradable and biocompatible polymer. The characterization of PHBHV coatings, native or doped with PEG, has been carried out at the nanoscale, in terms of wettability, topography and stiffness, and compared with PDMS coatings. With a hydrophobic character, a fairly rough surface topography and an indentation modulus in the GPa range, the efficiency of PHBHV coatings toward the adhesion of a bacterium and a diatom comes close to the performance of PDMS coating analogs. The strategy of modification of the hydrophobic PHA coating with the addition of PEG to

form an amphiphilic surface clearly enhances its anti-adhesive properties. With an increase in roughness and a more important stiffness than PDMS coatings, the adhesion results obtained both on a bacterium and a diaom highlight the importance of the wetting parameters and the amphiphilic character of the PHBV system in limiting the adhesion phenomenon. Owing to the existing chemical structures of PHA macromolecules, further studies should focus on the characterization of medium and long chain length PHA polymers to vary the crystallinity and test the anti-adhesive properties of this class of biopolymers.

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Disclosure statement

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